

Kishoreganj University
Department of Computer Science and Engineering
Mid Term Examination (1st session), 2024

Course Title: Artificial Intelligence

Course Code: CSE 3109

Session: 2021-22

Program: B.Sc.

Full Marks: 25

Total Time: 1 Hour

Answer the following questions.

Marks

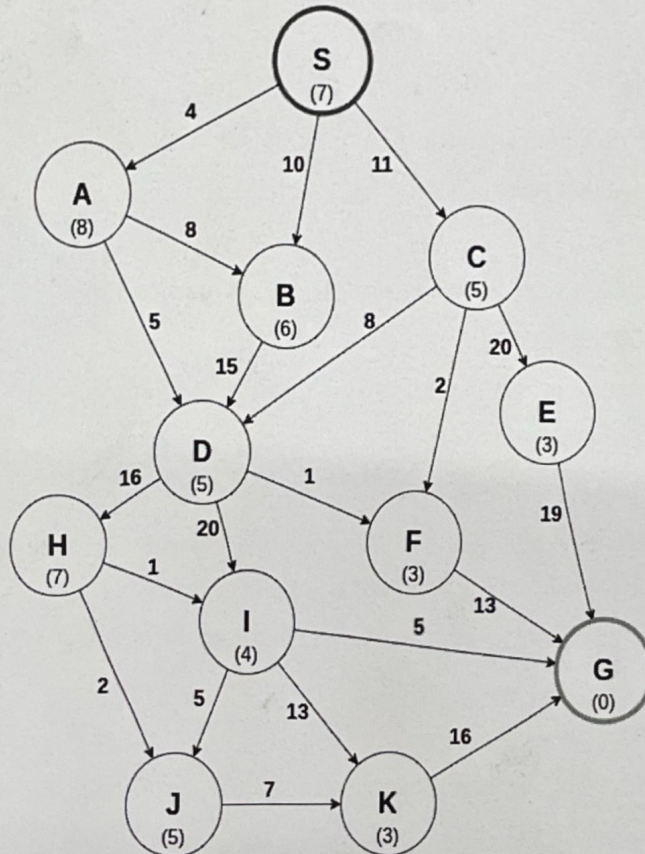
- ✓ 1. A Mars rover operates with delayed communication from Earth. Dust storms sometimes block its cameras, making parts of the terrain invisible. The rover uses previous data to guess possible obstacles. **[4+3+3]**
 - i. Classify the environment type for the rover (observable, deterministic, episodic, etc.) with justification.
 - ii. Which agent architecture best fits this scenario? Explain.
 - iii. Draw the architecture of the agent that is suitable for the given scenario.

2. An e-commerce company deploys a chat-based virtual customer support agent. **[2+4+3]**

Customers type their complaints, and the AI must respond naturally just like a human representative: answering queries, handling complaints, and adapting to different communication styles.

 - i. Which *view of AI* does this scenario represent?
 - ii. What essential capabilities must the AI possess in order to convincingly act like a human support agent?
 - iii. If the company upgrades the system with voice interaction (speech recognition) and visual understanding (analyzing product images or receipts), how would this shift the evaluation of the AI compared to the original test?

3. Consider the following directed graph. Node **S** is the source and node **G** is the destination. The values beside the edges represent the distances between nodes, while the values inside each node (enclosed in parentheses) represent the heuristic estimates of the straight-line distance from that node to the destination. [3+4+4]



- Apply Greedy Best First search to find shortest path from S to G.
- Apply A* search to find shortest path from S to G.
- When a heuristic is called admissible? Give some admissible heuristic functions that can be used for solving 8 puzzle problem. Which one is better in your opinion and why?